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Temporal variation in soft-sediment benthos

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Abstract: The existence of small-scale temporal variation in the distribution of the fauna of marine soft sediments has long been recognised. The implications of this variation have rarely been adequately addressed in studies of changes in the fauna over longer time-scales. Without knowledge of the presence or absence of variation at smaller scales, comparisons across longer time-scales must be confounded. Temporal variation in the distribution of the macrofauna of sediments in Botany Bay, New South Wales, Australia, is described using a nested, hierarchical sampling design. Significant variation was found at temporal scales from days to months. Implications of short-term variation for environmental sampling and monitoring, and the means of overcoming associated problems, are discussed. The conclusions of the present study are applicable to studies of other variables, such as pollutants.

Key words: Australia; Benthic macrofauna; Sampling design; Temporal variation

INTRODUCTION

Many studies have demonstrated the existence of small-scale temporal variation in variables related to marine soft sediments. For example, Stubblefield et al. (1977) and Warwick & Uncles (1980) described changes in the distribution of sediments in the New York Bight, USA, and the Severn Estuary, UK, respectively, in relation to storms or variation in the strength of water movements over the neap-spring tidal cycle. Livingston (1987) found variations in the abundance and composition of infauna at scales from weeks to years. Such faunal variation may be a result of changes in the sediments or may be due to migration, recruitment or to a variety of types of disturbance. As an example, Nichols & Thompson (1985) reported declines in the abundances of bivalves and total macrofauna coincident with erosion of sediment from the surface of a mud-flat over a period of a month. They also recorded almost total mortality of infauna caused by the deposition and decay of macroalgal material on the surface of the mud, the effects of which were evident for several months. The rapid rates of recolonization of sediments following defaunation (e.g., Santos & Simon, 1980) also indicate the likely importance of short-term variation in abundance.

Studies of the distribution and abundance of animals, including environmental im-

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pact studies, often involve comparisons through time. Small-scale temporal variation, however, has the potential to confound larger-scale comparisons (Underwood, 1991a,b). This is equivalent to the more widely recognized problem of the confounding effect of small-scale spatial variation on larger-scale comparisons, discussed by Green (1979), Hurlbert (1984), Andrew & Mapstone (1987) and Morrisey et al. (1992). Comparing the level of a variable in, say, 1 month with that in another based on 1 day of sampling in each month is the temporal equivalent of comparing one mudflat with another on the basis of one sample taken in each.

Consider a study of seasonal variation in numbers of a polychaete in some estuarine habitat. At each 3-month interval, a set of replicate samples is taken. Note that these samples are usually taken on the same day, or within a very short period. The variations among them are, therefore, *spatial* differences and do not provide any estimate of variability in time. If there are statistically significant differences among the means from each season, these are usually interpreted to reflect a seasonal cycle or pattern of difference. Yet, the differences could equally well be due to shorter-term variation, say from day to day, week to week or month to month (Underwood, 1989a,b). The only way to be sure that the temporal differences identified are really associated with the temporal scale claimed (in this case, from season to season) is to demonstrate that differences at smaller temporal scales are not as large.

This is analogous to the situation where spatially confounded samples are taken. For example, mean abundance of some organism in two areas, one experimental and one control, can be sampled by several replicates, compared statistically and shown to be different from one area to the other. It is not, however, appropriate to conclude that the experiment caused the difference (Hurlbert, 1984; Underwood, 1986). Any geographic or historic source of variation could have caused the observed difference. What are required are several replicated experimental and several replicated control areas, interspersed at some relevant spatial scale, to ensure there is no confounding.

So it is with temporal sampling. The mean of one set of replicates taken simultaneously or in a short period within one season may differ from the mean of another set in another season, but the difference may not be seasonal. Demonstration of seasonality requires demonstration that there is more variation between seasons than is found from time to time within seasons, just as demonstration of a treatment effect requires that there is more variation between treatment and controls than among replicate areas within a treatment.

Consideration of the diverse environmental factors that might potentially cause variation in faunal abundance over short time-scales would suggest that such fluctuations are common. Consequently, comparisons among samples collected at longer intervals will often be confounded by shorter-term variation. If, for example, variation in faunal abundance from one week to another is of a similar magnitude to the variation among months, then the size of differences among months, as estimated by sampling, will depend on which week within a particular month the samples were

collected. Published studies of human impact upon the marine environment, in which faunal abundances are compared before and after impact, provide many examples of this problem.

The confounding due to variation at small temporal scales, described above, occurs in addition to that caused by small-scale spatial variation (Peterson, 1975; Morrisey et al., 1992). The latter arises from the fact that samples will not necessarily come from precisely the same place at each time of sampling. As a consequence, any spatial heterogeneity in the fauna is likely to be interpreted as an effect of time. Adequate spatial replication of samples would have shown that, in fact, this variation was present within each sampling occasion. The importance of small-scale spatial variation for the interpretation of temporal trends in the distribution of heavy metal pollutants has been pointed out by Krumgalz et al. (1989).

Such problems of confounding can be overcome by the use of a sampling schedule incorporating nested, replicated sampling times at different temporal scales. In this manner, the contribution of each scale to the overall variation among samples can be estimated. The significance of variation at the scales of interest can be tested against smaller-scale variation. The time scales used (days, weeks, months, etc.) can be adapted to the aims of a particular study.

There are, in fact, many published studies in which samples were taken at regular intervals, such as fortnightly or monthly. Data from these studies could potentially be used to detect seasonal or yearly differences, in the manner described above, by treating individual sampling dates as replicates within each season or year, although they would not be randomly sampling a period of time. Regular temporal sampling can be problematic (e.g., Stewart-Oaten et al., 1986). Variation among seasons or years could then be tested against variation within each season or year, as estimated by the variation among sampling dates. This is rarely, if ever, done. Perhaps because time is obviously a continuous variable, people overlook the point that their samples are collected at particular times and are not necessarily typical of, nor adequately describe, the time-scale that they are taken to represent.

In the present study, we wished to determine the extent to which small-scale temporal variation may confound seasonal patterns in abundance of marine fauna. We examined variation in faunal abundance in soft sediments in Botany Bay, New South Wales, Australia, at a range of time-scales, from days to "seasons". Sampling was done according to a hierarchical, nested design, as described above. As far as we are aware, no previous studies have looked at time-scales in this way in order to unconfound comparisons over time. The results have relevance to studies of faunas of soft sediments elsewhere and also to the sampling of other marine habitats and other variables, such as pollutants.

METHODS

SAMPLING DESIGN

The sampling design incorporated two spatial scales; two plots, ≈4 m diameter and 10 m apart, were nested within each of two locations 1.5 km apart (Fig. 1). A previous study (Morrisey et al., 1992) demonstrated the existence of patchiness in the distribution of the benthic fauna in Botany Bay at scales of 1, 10 and 100 m, and 1 and 3.5 km. Use of replicate plots and locations in the present study was intended to provide greater generality to the results and to identify temporal–spatial interactions.

Each plot was sampled on each of 2 consecutive days in each of 2 consecutive weeks in each of 2 consecutive months in each of 2 consecutive "seasons" (March–April and June–July 1991). We considered that the days, weeks and months were representative of such time-scales and could be analysed as random, nested sources of variation in our hierarchical analysis. Three replicate cores (see below) were collected at each plot on each occasion. These were randomly placed, 1 m apart, within each plot.

METHODS OF SAMPLING

The positions of plots were marked with stakes on the sea-bed so that they could be relocated precisely on each sampling occasion. Replicate cores in each plot were almost certainly independent of the previously sampled cores. Depth of water at all the sites was ≈6 m. Samples were collected by divers using plastic core tubes (10 cm diameter, 10 cm deep). Pilot tests had shown that the precision of the estimates of faunal numbers obtained with this size of core was acceptable ($SE/\bar{x} = 0.1$; Elliott,

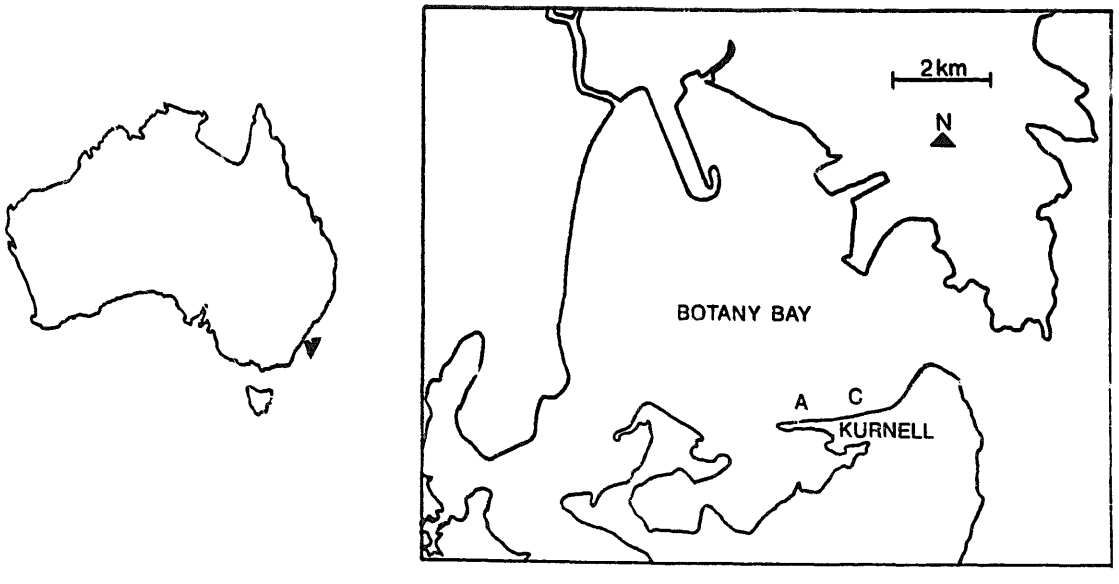


Fig. 1. Map of Botany Bay showing two locations sampled, Kurnell A and C.

1977). Numbers of animals collected with 20 cm-deep cores were no greater than with 10 cm-deep cores, hence we used the latter. Samples were fixed with 7% formalin and sieved through 0.5 mm. Identification of the fauna was to a relatively coarse level; families in the case of polychaetes, sub-orders or higher taxa for other groups.

Seven of the taxa in the samples were chosen as being consistently present and representing different relative abundances (see Table I). These, plus the total numbers of individual animals and the total numbers of taxa, were each analysed by four- and five-factor ANOVAs. The four-factor analyses included the four nested temporal scales (days, weeks, months and seasons), with each plot and each location analysed separately. The five-factor analysis included the four nested temporal scales and plots orthogonal to all of the time-scales. The two locations were analysed separately to simplify the demonstration of the problems for interpretation caused by inadequate temporal scales of sampling. All factors, spatial and temporal, were random. Homogeneity of variances was examined using Cochran's test and heterogeneity was removed by log e $(x + 1)$ or $(x + 1)^{0.5}$ transformations.

The mean-squares estimates for the five-factor analyses are shown in Table II. Note that differences among various time-scales are not testable unless interactions with spatial differences (between plots) are non-significant and can be eliminated from the linear model for the analysis (for full details of procedures, see Winer, 1971; Underwood, 1981). This is not a problem because where there are such interactions with plots,

TABLE I

Summary of four-factor ANOVAs of mean abundances at various time-scales in four plots in Botany Bay. Temporal scales at which significant variation exists among replicates are shown. Each plot at each location was analysed separately. –, no differences were significant; *, significance at $P<0.05$; **, at $P<0.01$.

Taxon	Kurnell A		Kurnell C	
	Plot A	Plot B	Plot A	Plot B
Total number of taxa	Seasons**	–	–	Weeks*
Total number of individuals	Seasons*	a	Weeks*	Days* Months**
Polychaetes	–	–	–	Days* Months*
Syllids	–	–	–	Days** Months**
Spionids	Seasons**	Seasons**	Days** Months**	Months**
Sabellids	–	Weeks* Seasons**	Days* Weeks* Seasons**	Months**
Amphipods	Months*	–	Weeks**	Months**
Cumaceans	Months**	Months**	Days* Weeks*	Months**
Anthurans	–	Seasons*	–	–

[a] Variances could not be homogenized by transformation; analysis not summarised here.

TABLE II

Mean squares estimates (Winer, 1971; Underwood, 1981) and F ratios for a five-factor ANOVA with randomly chosen plots orthogonal to a series of random, nested time-scales; pooling of non-significant sources of variation to create F ratio tests for temporal differences is explained in Winer (1971) and Underwood (1981).

Source of variation	df	Mean squares estimates	F vs.
Season	1	$\sigma^2_e+3\sigma^2_{PD(W(M(S)))}+6\sigma^2_{PW(M(S))}+12\sigma^2_{PM(S)}+24\sigma^2_{PS}+6\sigma^2_{D(W(M(S)))}+12\sigma^2_{W(M(S))}+24\sigma^2_{M(S)}+48\sigma^2_S$	No test
Month (S)	2	$\sigma^2_e+3\sigma^2_{PD(W(M(S)))}+6\sigma^2_{PW(M(S))}+12\sigma^2_{PM(S)}+6\sigma^2_{D(W(M(S)))}+12\sigma^2_{W(M(S))}+24\sigma^2_{M(S)}$	No test
Week (M (S))	4	$\sigma^2_e+3\sigma^2_{PD(W(M(S)))}+6\sigma^2_{PW(M(S))}+6\sigma^2_{D(W(M(S)))}+12\sigma^2_{W(M(S))}$	No test
Day (W (M (S)))	8	$\sigma^2_e+3\sigma^2_{PD(W(M(S)))}+6\sigma^2_{D(W(M(S)))}$	P × D (W (M (S)))
Plot	1	$\sigma^2_e+3\sigma^2_{PD(W(M(S)))}+6\sigma^2_{PW(M(S))}+12\sigma^2_{PM(S)}+24\sigma^2_{PS}+48\sigma^2_P$	P × S
P × S	1	$\sigma^2_e+3\sigma^2_{PD(W(M(S)))}+6\sigma^2_{PW(M(S))}+12\sigma^2_{PM(S)}+24\sigma^2_{PS}$	P × M (S)
P × M (S)	2	$\sigma^2_e+3\sigma^2_{PD(W(M(S)))}+6\sigma^2_{PW(M(S))}+12\sigma^2_{PM(S)}$	P × W (M (S))
P × W (M (S))	4	$\sigma^2_e+3\sigma^2_{PD(W(M(S)))}+6\sigma^2_{PW(M(S))}$	P × D (W (M (S)))
P × D (W (M (S)))	8	$\sigma^2_e+3\sigma^2_{PD(W(M(S)))}$	Residual
Residual	64	σ^2_e	

temporal patterns must be interpreted on a plot-by-plot basis using multiple comparisons within each plot (Underwood, 1981). Therefore, there is no purpose in attempting tests of the time-scales themselves unless there are no interactions with plots.

Results

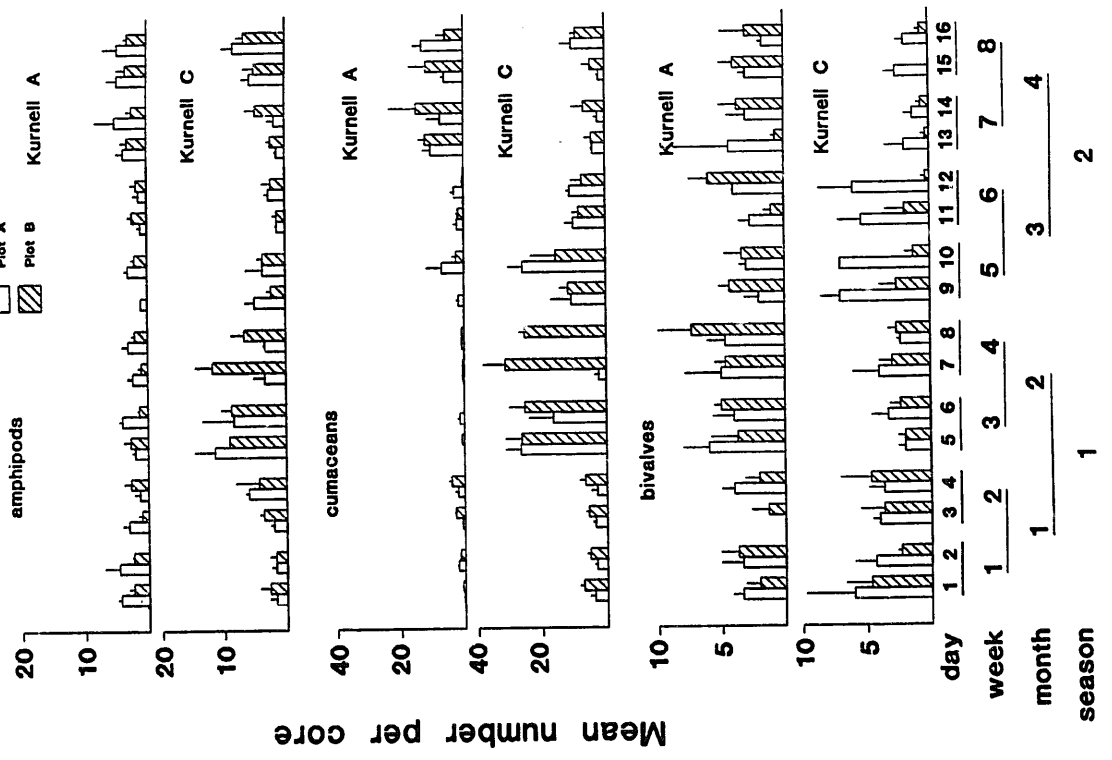
Among the data for the various taxa analysed, significant differences were found at all temporal scales examined (Tables I, III, Fig. 2). This is shown most clearly in the analyses in which plots were examined separately, omitting temporal–spatial interactions from the analyses (Table I). For example, cumaceans at Kurnell C, Plot A varied in mean abundance from day to day and week to week. Sabellids in the same plot varied among days, weeks and seasons. Spionids at Kurnell C, Plot B varied from month to month, while at Kurnell A, Plot A they varied only from season to season. Anthurans at Kurnell C showed no significant variation at any of the time-scales examined. No taxon showed the variation at the same temporal scales at both locations nor, in most cases, at the two plots within a location. Spionids and cumaceans at Kurnell A, however, did vary at the same scale at both plots (seasons in the case of spionids, months in the case of cumaceans).

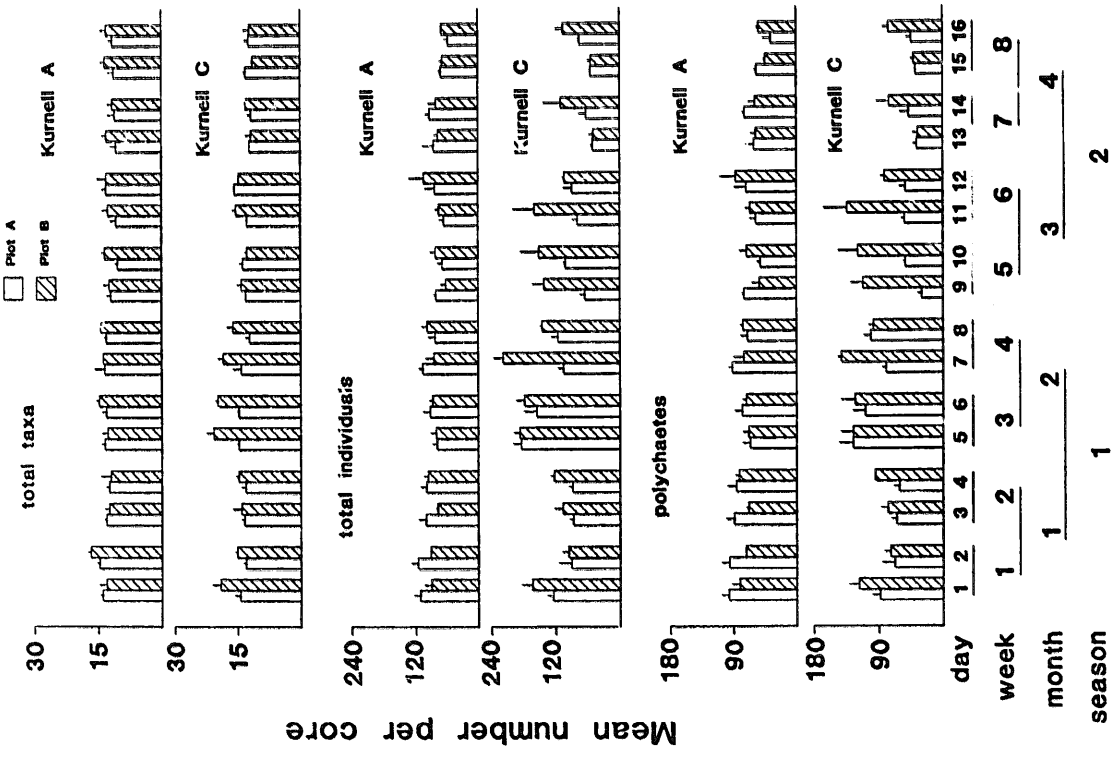
When the two plots at each location were considered in the same analyses (Table III), there were several significant differences between replicate plots and also significant plot–time interactions. For example, the numbers of individual animals showed a significant difference between plots at Kurnell C and the numbers of polychaetes at the same location showed a significant plot × month interaction (Table III). Differences in total numbers of individuals among plots were particularly marked at Kurnell C (Fig. 3). Temporal variation was also more pronounced at this location, although the mid-points of the fluctuations of all four plots were fairly similar. The existence of time–plot interactions demonstrated that the temporal patterns of variation in mean abundances of some organisms were not the same at the two plots within a location. Had locations been incorporated into the analyses as a factor, significant time–location interactions would probably also have been found.

Discussion

The results show variation in the fauna over all time-scales examined, from days to several months. Variation is also likely over shorter or longer periods (e.g., tidal cycles, including rhythms of migration, year-to-year variations, correlations with climatic changes over decades or centuries, El Niño-type hydrographic events; see references in Barry & Dayton, 1991).

The differences found between replicate plots with some locations show that agents of temporal change in faunal associations which act over broad spatial scales, such as


Plot A
Plot B
amphipods
cumaceans
bivalves
Kurnell A
Kurnell C
Mean number per core
day
week
month
season


Plot A
Plot B
total taxa
total individuals
polychaetes
Kurnell A
Kurnell C
Mean number per core
day
week
month
season

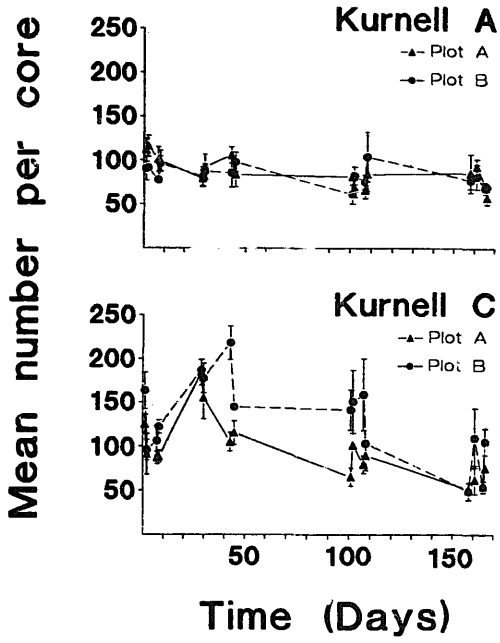


Fig. 3. Mean (± SE) numbers of individuals of all taxa in each plot ($n = 3$) at each time of sampling. Time is shown as number of days after first time of sampling (Day 1).

large-scale water movements, tidal and seasonal changes, etc., are not the only ones that are important. Factors acting at small spatial scales can apparently also alter the time course of faunal associations. Such factors might include localized water movements and their associated variables (supply of food, oxygen, larvae, pollutants, etc.) and disturbances by, for example, the activities of predators (Van Blaricom, 1982; Oliver & Slattery, 1985; Thrush, 1986; Hall et al., 1991). The correlation between the magnitudes of spatial and temporal scales pointed out by Barry & Dayton (1991) may apply in this context. Thus, short-term variations would result mainly from factors acting at small spatial scales while factors acting at large spatial scales would be responsible for longer-term changes.

The results of the present study have important consequences for studies of temporal variation over larger scales. These include the variety of before/after, control/impact ("BACI") designs for environmental impact studies (reviewed in Underwood, 1991a). First, as described in the Introduction, lack of replication within each time period being compared will confound comparisons among periods. As an illustration of such confounding, consider the results obtained for cumaceans at Kurnell C, Plot A (Table I), in which significant variation was found at scales of days and weeks, but not at longer scales. Any attempt to compare, say, different seasons based on sampling on a single day or even over a single week in each season would be confounded by day-to-day and

Fig. 2. Mean (± SE) numbers of taxa or animals per core ($n = 3$) in each plot; there are two plots in each of two locations. Samples were collected on each of 2 days in each of 2 weeks in each of 2 months in each of 2 "seasons" (3 months apart).

TABLE III

Summaries of ANOVAs for selected taxa at Kurnell A and C; S, season; M, month; W, week; D, day; P, plot. [1] Indicates data were transformed to $\log_e(x+1)$; pooling of non-significant interactions was done at $P = 0.25$ (Winer, 1971; Underwood, 1981); Pld, source of variation pooled from mean squares with appropriate superscripts and with corresponding df. For example, for Kurnell A, (a) No. taxa, factor "season" is tested against pooled mean squares for months and weeks with 1 and 6 df. To simplify table, F and P values have been omitted for those factors and interactions where $P > 0.25$.

Kurnell A	df	(a) No. taxa				(c) No. individuals				(c) No. polychaetes				(d) No. amphipods[1]				(e) No. cumaceans[1]				(f) No. bivalves[1]			
Source		MS	F	P	F vs.	MS	F	P	F vs.	MS	F	P	F vs.	MS	F	P	F vs.	MS	F	P	F vs.	MS	F	P	F vs.
S	1	49.59	9.05	<0.05	Pld^{c}	6700.06	9.58	<0.05	Pld^{c}	8066.66	6.20	>0.05	M	0.01	0.00	>0.25	M	21.77	1.71	>0.25	M	0.23	0.08	>0.25	M
M	2	0.26^{c}				505.69^{c}				1301.78	2.31	>0.05	W	2.20	7.47	<0.01	Pid^{b}	12.76	15.01	<0.05	Pld^{b}	2.81	3.88	<0.05	Pld^{b}
W	4	8.09^{c}	1.78	>0.05	Pld^{b}	795.80^{c}	1.96	>0.05	Pld^{b}	563.91	1.82	>0.05	Pld^{b}	0.32^{b}				0.29				0.48^{b}			
D	8	2.93^{b}				310.63^{b}				221.51^{b}				0.34^{b}				0.57				0.85^{b}	2.04	>0.05	Pld^{a}
P	1	19.26				425.06				468.16				1.19	4.12	<0.05	Pld^{a}	0.11^{b}				0.03^{a}			
P × S	1	7.59	1.61	>0.05	Pld^{a}	828.38	1.99	>0.05	Pld^{a}	737.06	2.31	>0.05	Pld^{a}	0.10^{a,b}				1.16^{b}				0.04^{a}			
P × M	2	0.76^{a,b}				484.38^{a,b}				339.61^{a,b}				0.06^{a,b}				1.05^{b}	2.41	>0.05	Pld^{a}	0.07^{a}			
P × W	4	0.60^{a,b}				229.88^{a,b}				143.66^{a,b}				0.30^{a,b}				0.39^{a}				0.20^{a}			
P × D	8	2.80^{a,b}				301.78^{a,b}				349.14^{a,b}				0.26^{a,b}				0.46^{a}	1.42	>0.05	Res.	0.44^{a}			
Residual	64	5.33^{a,b}				440.18^{a,b}				324.94^{a,b}				0.30^{a,b}				0.32				0.45^{a}			

Kurnell C	df	(a) No. taxa				(c) No. individuals				(c) No. polychaetes				(d) No. amphipods[1]				(e) No. cumaceans[1]				(f) No. bivalves[1]			
Source		MS	F	P	F vs.	MS	F	P	F vs.	MS	F	P	F vs.	MS	F	P	F vs.	MS	F	P	F vs.	MS	F	P	F vs.
S	1	133.01	3.53	>0.05	M	47570.50	1.85	>0.05	M	25090.69	2.15	>0.05	M	1.26	0.24	>0.25	M	0.16	0.02	>0.25	M	3.02			
M	2	37.68	2.64	>0.05	W	25712.50	10.09	<0.01	Pld^{b}	11666.41				5.28	2.61	>0.05	W	8.28	3.41	>0.05	W	3.09	8.96	<0.005	Pld^{b}
W	4	14.28	2.76	>0.05	D	1953.84^{b}				607.16				2.02	5.69	<0.005	Pld^{b}	2.43				0.22^{b}			
D	8	5.18	1.51	>0.05	Pld^{a}	2844.53^{b}	1.82	>0.05	Pld^{a}	1255.14	1.67	>0.05	Pld^{a}	0.18^{b}				0.83	1.98	>0.05	Pld^{a}	0.08^{b}			
P	1	78.84				29155.50	18.60	<0.01	Pld^{a}	20768.19				0.40				7.10				6.44			
P × S	1	71.76	6.72	>0.10	P × M	90.13^{a}				2562.63				0.23				6.57	2.88	>0.05	Pld^{b}	4.06	10.73	<0.005	Pld^{a}
P × M	2	10.68	3.10	>0.05	Pld^{a}	2430.06^{a}				3680.41	4.91	<0.05	Pld^{a}	0.59	1.58	>0.05	Pld^{a}	1.70^{b}				0.42^{a,b}			
P × W	4	3.53^{a}				1423.81^{a}				355.47^{a}				0.46^{a,b}				2.57^{b}	6.12	<0.005	Pld^{a}	0.10^{a,b}			
P × D	8	2.93^{a}				1607.75^{a}	1.82	>0.05	Res.	946.88^{a}	1.70	>0.10	Res.	0.36^{a,b}				0.42^{a}				0.12^{a,b}			
Residual	64	3.50^{a}				882.73				556.70				0.37^{a,b}				0.42^{a}				0.43^{a,b}			

week-to-week variation within each season. Conversely, because there is large variability at shorter time-scales, longer-term temporal patterns will not be detected without considerable replication at shorter periodicity. Statistical tests for differences in mean abundance at large temporal scales will not be powerful without large numbers of replicates where there are shorter-term fluctuations in abundances.

The second implication relevant to studies of patterns of temporal change is that differences between two places in their time courses of variation can clearly occur independently of any human activity. It is not justified to assume that temporal patterns observed at one place will occur at other places, even as close as 10 m. Therefore, demonstration of a change in the pattern of variation between a putatively impacted and a single control site coincident with a disturbance supposedly causing an impact is not sufficient to ascribe causality. This illustrates the need for multiple control sites in such studies, as described by Underwood (1991b, 1992). With replicated control sites, an impact will be detectable as a change in the difference between the pattern of temporal variation in abundances at the impacted site and the average variation of the several control sites.

In conclusion, the results of this study reveal the need for clustered, nested sampling times to be incorporated into studies of temporal variation in order to unconfound effects of short-term temporal variation. With respect to adequate replication in published studies, those involving comparisons over time are often inadequate because they do not incorporate any temporal replication at all. In contrast, in spatial studies, samples are usually replicated, even if at inappropriate spatial scales (Morrisey et al., 1992). Published studies of temporal variation usually treat each time of sampling separately, rather than as replicates, or they implicitly consider spatially scattered replicates at one time to be appropriate for estimating temporal variability. Thus, when comparisons among larger scales are made, any potential variation within them is ignored. As mentioned in the Introduction, it may be possible in some cases to analyse the results of such studies using sampling times as replicates within larger time-scales. These sampling times are usually spaced at equal intervals, rather than being in randomly spaced clusters within the larger time-scales, so that chance coincidence with environmental cycles may cause inadequate estimation of temporal variation (Stewart-Oaten et al., 1986). Because spacing is often not exactly even, this may not be a problem. Nevertheless, in general, temporal replication should be provided by random temporal sampling. For example, in a seasonal study at 3-month intervals, several temporally replicated samples should be taken within each season. Better still, seasons might be defined to have disjunct boundaries. Thus, each of the seasons could be considered to be two months long with a month between seasons. Several replicate periods of sampling should then be scattered within each season, to provide appropriate replication within seasons. In this way, distinctions among seasons, if they exist, will be identified without confounding. By having a "buffer" between seasons, any differences among seasons will not be blurred by similarities as one season gives way to another.

Finally, while the results described here have concerned the macrofauna of soft sediments, the conclusions drawn are likely to be equally applicable to other environmental variables, such as the distribution of sediments and pollutants. They are also relevant to other habitats, organisms and problems.

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